

DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression

DeepSeek-AI



Paper



Code

Problem

Long-context agents are increasingly input-heavy.

Expensive prefill and large KV caches strain accelerator memory, persistent storage, and transfer bandwidth, limiting economical deployment of million-token contexts.

Motivation

DESIGN PRINCIPLE

Share global context. Rebuild local state.

Treat **global KV** and **sliding-window state** differently: share and quantize the former; reconstruct the latter when needed.

Co-designing architecture and deployment reduces both serving and prefix-cache costs for repeated, million-token inputs.

GLOBAL CONTEXT

Long-lived KV dominates memory as the prompt grows.

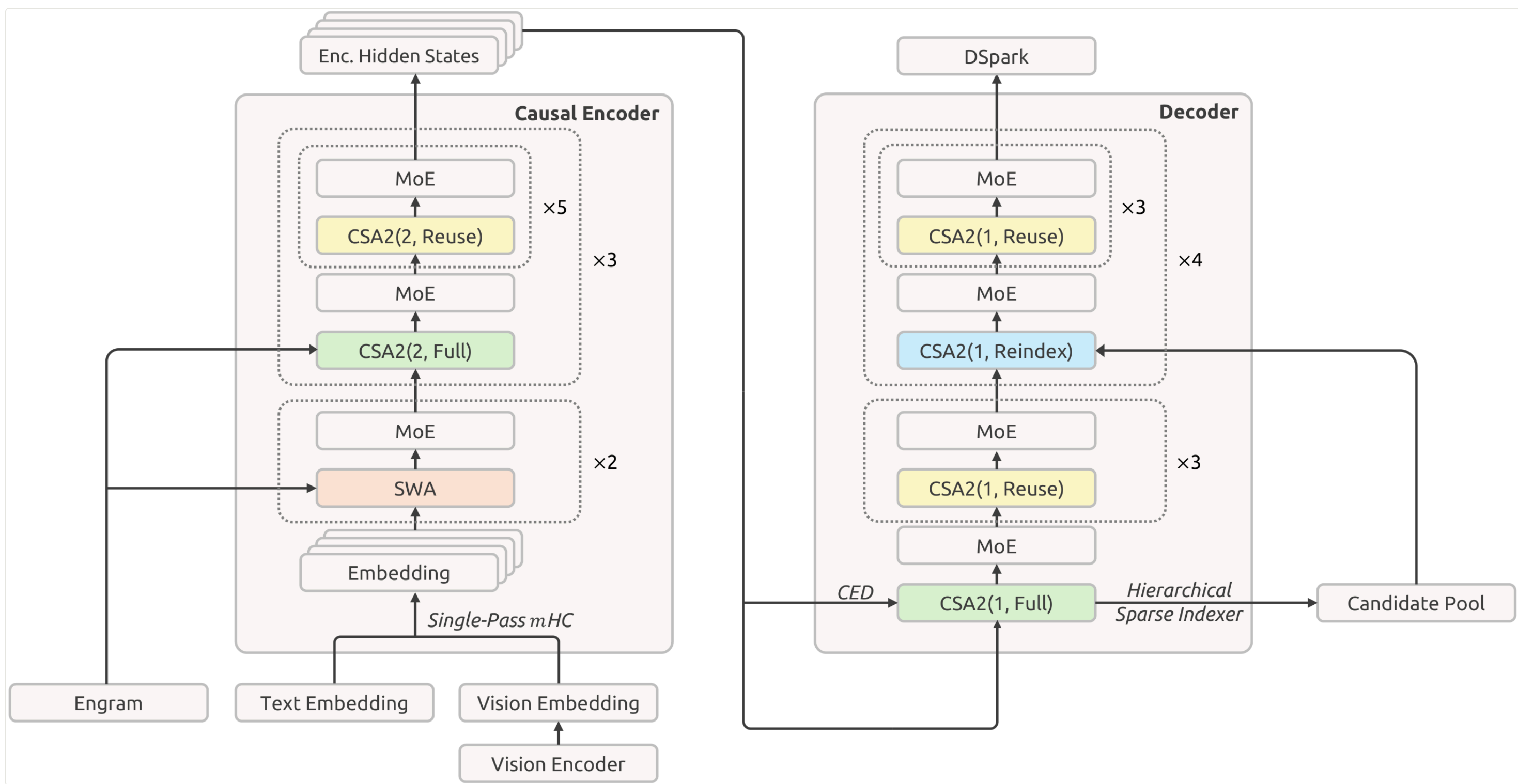
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LOCAL STATE

Recent tokens can recover short attention windows.

Exact local replay may span layers $\times$ window size. Smaller effective receptive fields make **bounded replay** practical.

Method



Architecture: 20 encoder + 20 decoder layers.

A Causal encoder-decoder

PREFILL

Project decoder global KV from the **final encoder states**, enabling most long-prompt prefill to stop halfway through the network.

$$C_l = H_{L/2} W_l^{KV}$$
$$Z_l = H_{L/2} W_l^Z, \quad l > L/2$$

C : KV entries · Z : compression weights

B Share KV and indices

CSA2

MODE	GLOBAL KV	TOP-K
Full	New	New
Reindex	Shared	New
Reuse	Shared	Shared

Layer-specific: every mode computes its own query and local sliding-window KV.

C Bound the search

INDEXING

- The first decoder Full layer scores the global context and ranks blocks.
- Collect a shared candidate pool for later indexers.
- Reindex layers select fresh Top-512 entries within this pool.

The first indexer still scans the full context. Up to **2,048** blocks $\times$ 8 positions form a **16,384-position** candidate pool. Training and inference use the same restriction.

D Store less, replay locally

CACHE

FP4 main KV

128-token replay

Store main KV at **four-bit precision**. Reconstruct sliding-window attention from a short replay of the most recent **128 tokens**.

FP4 E2M1 stores main KV with one E4M3 scale per 16 channels. Quantization follows RoPE; values are dequantized before attention. Quantization-aware training is introduced during post-training.

SWA KV stays FP8 because local attention is more sensitive to quantization.

Persist global KV and omit SWA KV from persistent storage. Replay is approximate and depends on the cache-hit position; the authors report negligible response-quality degradation.

Key Results

BENCHMARK	V4-FLASH	V4.1-FLASH
DeepSWE	54.4	74.2
TB 2.1	82.7	90.6
Automation	37.7	54.8
Agents' Last Exam	25.2	31.8

Table 3 · Max reasoning effort. Scores are percentages; DeepSWE uses mini-SWE, and TB 2.1 uses DeepSeek Harness Minimal.

Headline Numbers

890

bytes / token · global KV
About 1/4 of V4-Flash global KV

4× smaller global KV

8× smaller persistent KV

Takeaway

Smaller caches, stronger agents.

Joint model and serving design makes million-token agents more efficient while improving reported agent performance.

Benchmark gains depend on reasoning effort and scaffold. The memory ratios do not measure an end-to-end serving-cost reduction.